NAO VI Guide

Basic Instructions

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For:

The NEST Lab, University of British Columbia

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Contents

Introduction 3
Setup 3
Interaction 4
Rest Mode 6
Shutting Down 6
Limitations 6
  Battery Life 7
  Environment 7
  Interaction 7
  Mobility 7
Dos and Don’ts 8
  Do 8
  Don’t 8
Troubleshooting 8

Figures

Figure 1: Robot in crouch position 3
Figure 2: NAO system flowchart 4

Introduction

This document provides explanations and details involving the NAO robotics, Social Robot project accompanying the video documentation also provided. The purpose of this document is to describe the limitations, dos and don’ts, best practices, and any other pertinent or helpful information that can be included outside the videos regarding the operation of the NAO robot. Each section of the report is detailed with information that will clarify any confusion surrounding the operation of the NAO robot. These explanations aim to serve as reference for development, debugging, and demonstration within the NEST (Neuroscience Engagement Smart Tech) Lab.

Setup

Upon removing the robot from the box, place it in the rest position (below). The robot should be sitting on its heels with legs bent and hands resting on knees. The robot should be self-balanced in this position and will return to this position when it enters rest mode (double tap chest button). Holding the center button will turn the robot on. Please wait up to 2 minutes for the robot to fully wake up, at which point the robot will stand up and begin searching for people to engage with.

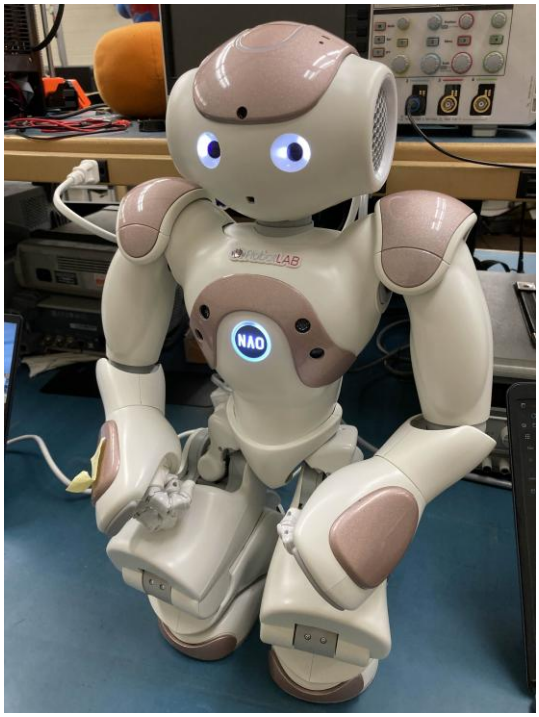

Figure 1: Robot in crouch position

Interaction

To interact with the NAO, the participant must stand close to the robot until the eyes turn blue; this indicates the robot is now listening to the person that is being looked at.

The NAO as we have programmed it has 3 main behaviors: Dance, Joke, Stretch. They can each be entered through voice commands. These commands are best done as single words: “dance”, “joke”, “stretch”. All three behaviors can be exited manually at any time by tapping either of the robot’s toe sensors. Please find the chart below to see the flow of the behaviors.

Additionally, when the robot is not performing one of these behaviors, it will respond to many other voice prompts. Please see the list below; these are the commands that have been tested and confirmed to work. A full list, including the commands that didn’t seem to work can be found here: http://doc.aldebaran.com/2-1/nao/basic_channel_conversation.html

The robot’s feet LEDs will show the color indicated on the flowchart. This allows the user to visually identify the behavior the robot is currently in, with no color on the feet representing the robot being in autonomous mode.

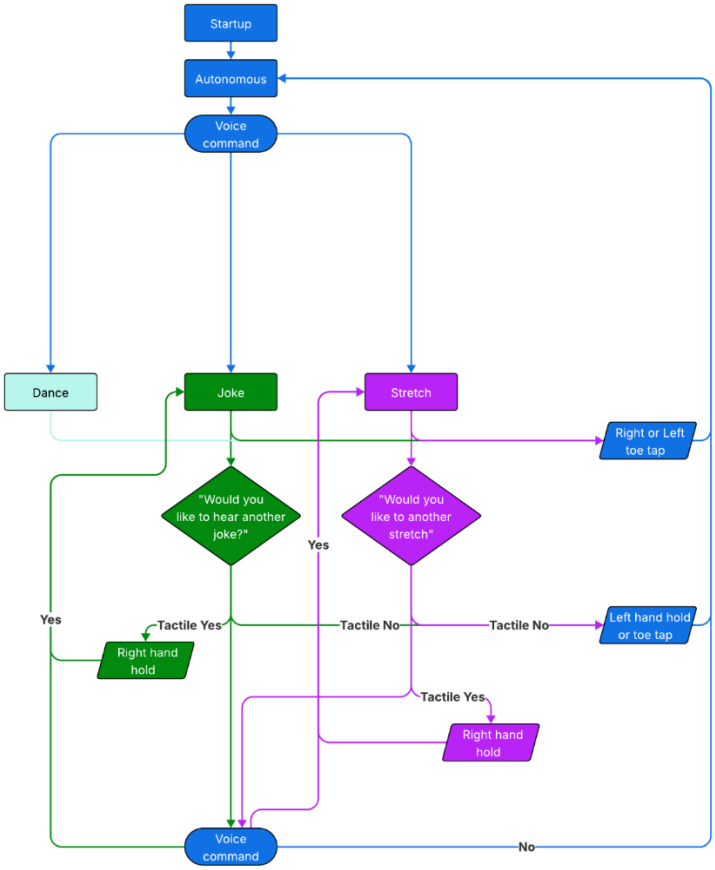

Figure 2: NAO system flowchart

Introduction

- How are you
- Can you say goodbye

Volume control

- Speak softer
- Speak louder

Movement (be careful)

- Can you stand up
- Can you sit down
- Can you move your head
- Can you turn your head (left/right)
- Can you look up
- Stop looking at me
- Can you stretch your (left/right/both) arm
- Can you raise your (left/right/both) arm
- Make your hands move
- Can you open your hands
- Can you close your hands
- Can you put your hands behind
- Hands up!

About the robot

- Introduce yourself
- How tall are you
- How much do you weigh
- What color are you
- Are you kind
- Are you cool
- Are you intelligent
- Can I trust you
- Are you human
- Are you alive
- Do you eat
- Can you talk
- Can you see me
- Can you hear me
- Can you understand me

Sing a song (poorly)

- Sing me a song

Don't do this one (very boring)

- Do you know the laws of robotics

Mostly Aldebaran information

- Why is your name NAO
- What is your name (NAO)
- What is your nickname (Chip)
- Who made you
- Where were you made
- Do you have a family
- Do you have friends
- Can we be friends

Rest Mode

The robot can be put into rest mode by tapping the center chest button twice. It will let out a noise and then enter the crouch position. At this point its actuators are no longer active and programs are suspended. Tapping the chest button another two times brings the robot out of rest mode and back to normal operation.

Think of the rest mode as a standby mode, this is where it should be whenever the robot is not currently being interacted with but needs to be ready to go. Exiting rest mode takes only a few seconds, and when it is in rest mode you don't need to worry that the robot will do anything silly to hurt itself.

Note: Please remember to always enter rest mode before moving the robot. Moving the robot without entering rest mode could result in damage to the motors.

Shutting Down

When finished with the robot or if experiencing issues with the performance of the robot, please perform a shutdown. This can be done by holding the chest button down for 3 seconds. The robot will make a noise to indicate it is beginning the shutdown procedure and begin moving to the crouching rest position. It is important that you release the robot and allow its motors to shut down gracefully. Once the NAO robot is shut down, please return it to the box, ensuring the fingers are closed and will not be damaged by the lid being placed on it.

Should the NAO fail to shut down from the 3 second chest hold the robot can be forced to shut down. This is done by holding the chest button for about 8 seconds. Please be aware that all of the motors will shut down instantly and the robot will fall over if not properly supported.

Updating the Jokes and Other Videos

Please see the video included on the robot computer to find a video showing the full process of updating the joke list within the robot. In the same place you will be able to find videos showing the setup of the robot on a hotspot, as well as a video showing the flow of behaviors and how to interact with them.

Limitations

In this section you can find the limitations of the robot we expect to impact your tests.

Battery Life

You can find this information in the NAO booklet, but it is here for ease of access

- Battery life can last from 45 minutes – 2 hours
- It takes 2 hours to charge the battery to full

We find you can expect about 60 minutes from the robot before performance declines.

Environment

We assume the NAO robot requires the same environmental accommodation as all the other robots. These primarily include the following:

- No sunlight directly in its "eyes"
- Keep the surroundings as quiet as possible
- As few people as possible around the robot to not overwhelm it
- Keep the robot away from heat sources. The electronics in its head generate a substantial amount of heat and could be prone to overheating.

Interaction

After running for too long, the NAO robot may begin to heat up. At this point the performance of the robot will decrease, and it may begin behaving erratically. We recommend running the robot and interacting with it for no more than an hour at a time, with a full shutdown between each session.

Mobility

We don't have the NAO robot setup to move about the environment, so it should not walk around anywhere. However, the robot does spin when performing the macarena dance and the leg stretches; please be mindful of these movements and ensure the robot stays centered in the table. Also, if the robot is asked to sit or lay down it will move in unpredicted ways. Be ready to catch it if it is near the edge of the table and remember to always handle it by its torso to avoid risk of pinching. Honestly, it is best practice to not ask it to lay or sit down unless the robot is in a completely safe position.

Should the robot fall, it will attempt to stand up on its own. This is a dangerous time. If safe to do so, give the robot center button a double tap to enter "rest" mode. This will disable the joint actuators, at which point you can manually move it back to the crouch position and double tap the center button again to re-enter standard operation.

Dos and Don’ts

Do

- Hold the NAO robot by the waist when moving around
- Move the NAO when its motors are off (crouching or sitting position)
- Use soft dry cloth or cleaning wipes for cleaning
- Ensure hands of NAO are closed when placed in box and lined with the casing
- Support the feet while carrying the robot

Don’t

- Put your hands near joints or between limbs as they could be pinched
- Pull NAO robot’s head, arms, or legs, you could damage the joints
- Pull on the NAO robot’s fingers as they can come off
- Block the sensors and joints
- Put any clothing accessory on the robot
- Put makeup or paint on the robot
- Cover the head, mainly the air vent on the back of the robot’s head
- Use water for cleaning, get NAO wet or use wet hands
- Alcohol or ammonia-based products for cleaning
- Oil the joints
- Service the robot yourself

Troubleshooting

If the NAO does not recognize faces and begin listening to people (blue eyes) – a simple fix is to send the robot to rest mode (double tap chest), bring it back out (double tap chest) and place your hand on its head sensors for 3-10 seconds. This will usually get it back to interacting properly. If it doesn't work, a full power cycle is required (takes about 5 minutes).

If the NAO is speaking over itself and providing nonsense responses consistently – the robot is probably too hot or has been running too long. Giving the robot a break by turning it off for a while is the way to go.

If the NAO is not listening or refuses to end a behavior – tap its toes to exit the behavior, hold your hand on the robot’s head sensors for 3-10 seconds to stop any other programs. If this does not work send it to rest mode for a few seconds (double tap chest) and then bring it back.

If the NAO says it has an error – turn the robot off and on again by holding the center button for about 3 seconds.

If the NAO falls over – the robot has a protocol for standing back up. If it seems to be struggling unduly or might fall off the table carefully try to put it into rest mode (double tap chest). Be very careful not to have your fingers pinched or resist the motors too much as either one will cause damage to either your hand or the robot.

If the NAO will not shut down after holding the chest button for about 3 seconds – the robot can be forced to shut down. This is done by holding the chest button for about 8 seconds. Please be aware that all of the motors will shut down instantly and the robot will fall over if not properly supported.

If a finger breaks – there are no replacement parts available. Ensure all wiring is inside the finger and will not be pinched by movement and then tape the finger closed.